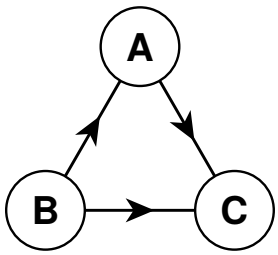
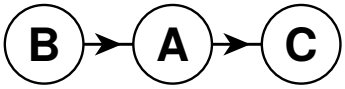
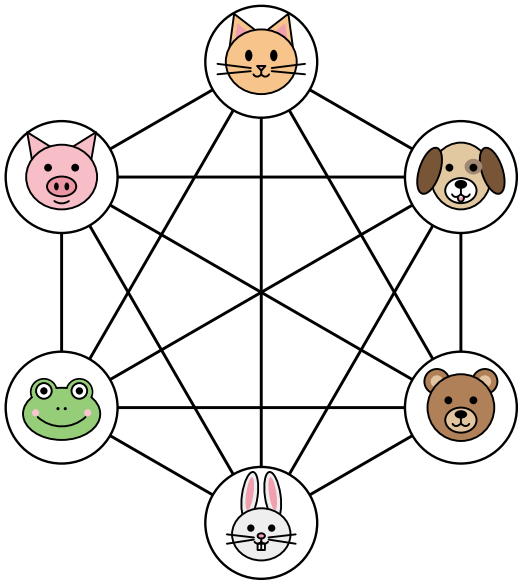
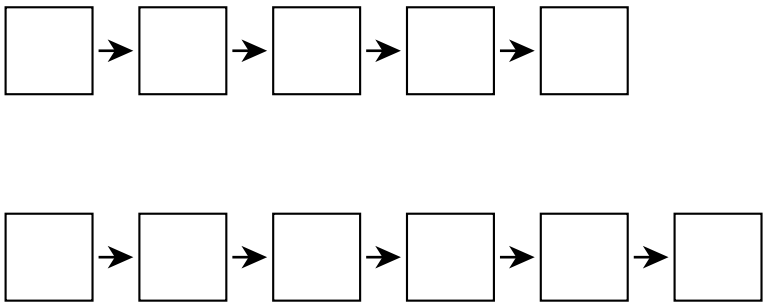
Every two players play one game, and one of them wins. An arrow points from the winner to the loser. A line-up is a row of all the players in which each player beat the next one.



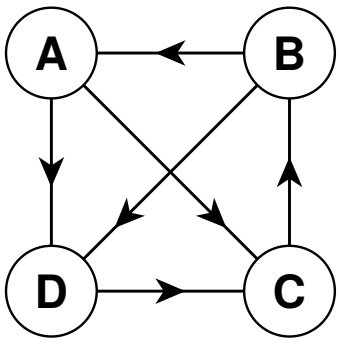


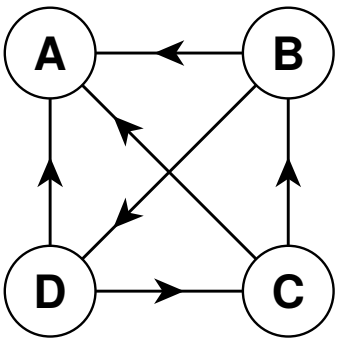
Problem 1: Play a tournament with the cat, dog, bear, rabbit and frog cards, deciding each game by rock-paper-scissors. Draw the arrows on the picture and find a line-up. Then the pig joins and plays the other five. Fit the pig into your line-up without changing the order of the others.

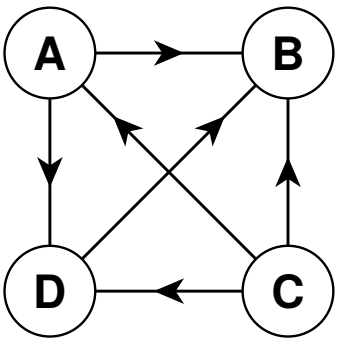


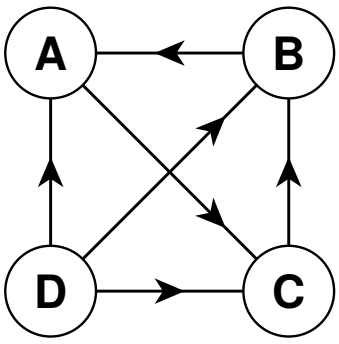


Problem 2: Find every line-up of each tournament.

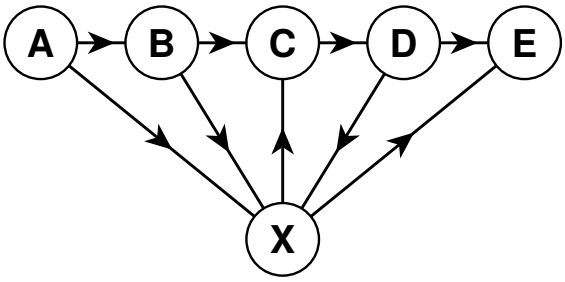


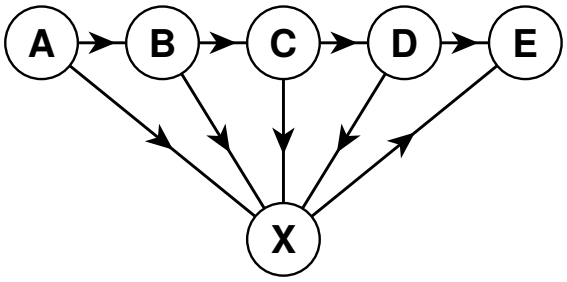


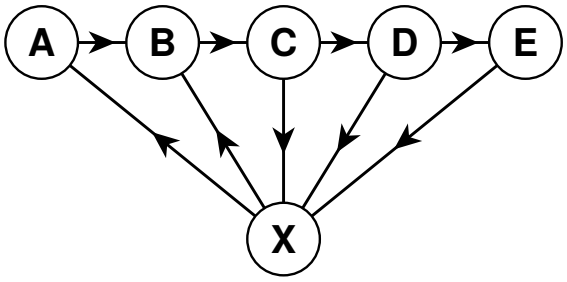




Problem 3: Player X plays each player in the line-up A, B, C, D, E. In each picture, mark every place where X can join the line-up without changing the order of the others. Can X always join somewhere, whatever the results? Explain.

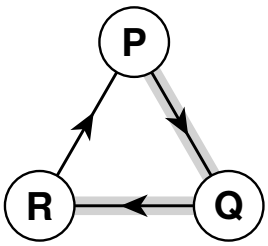


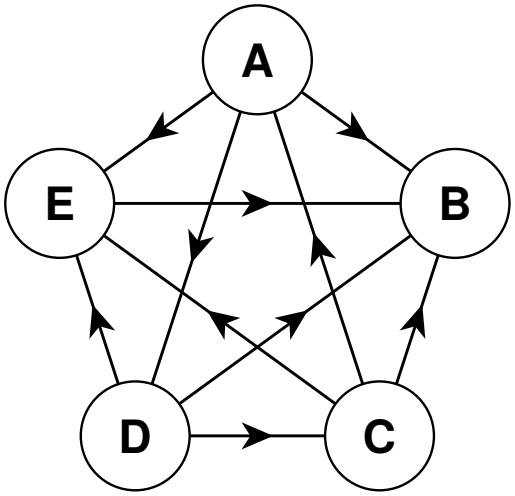


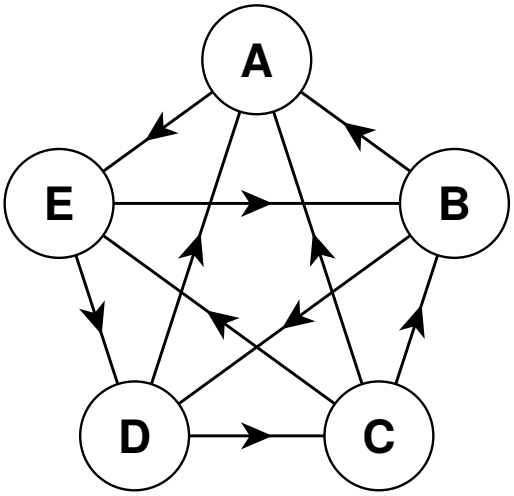


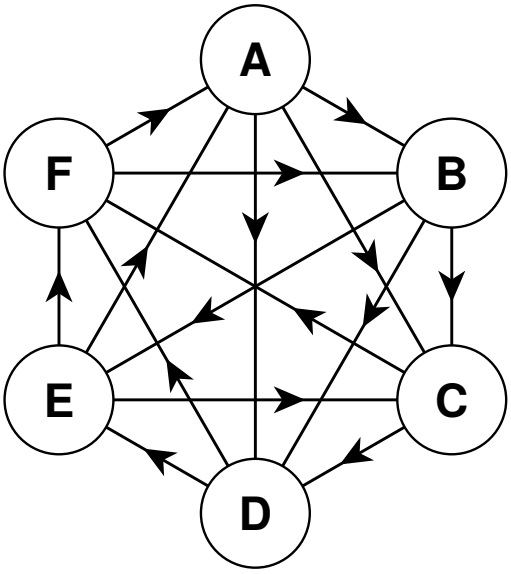
Problem 4: Does every tournament have a line-up, however many players it has? Explain.

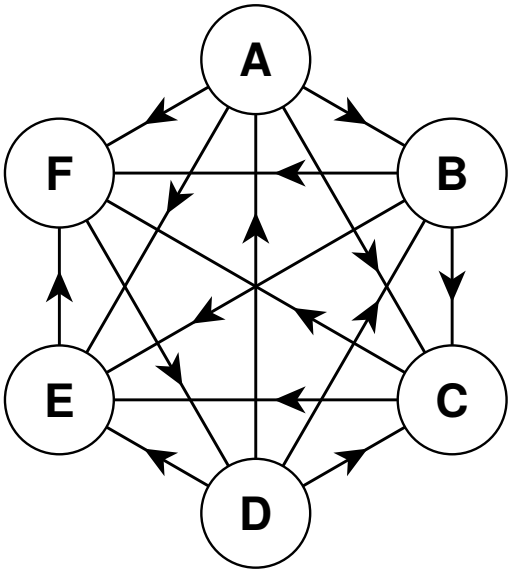
Problem 5: A player is a king if you can get from that player to every other player by following one arrow or two arrows. Here P gets to Q by one arrow and to R by two arrows. Find all the kings in each tournament.





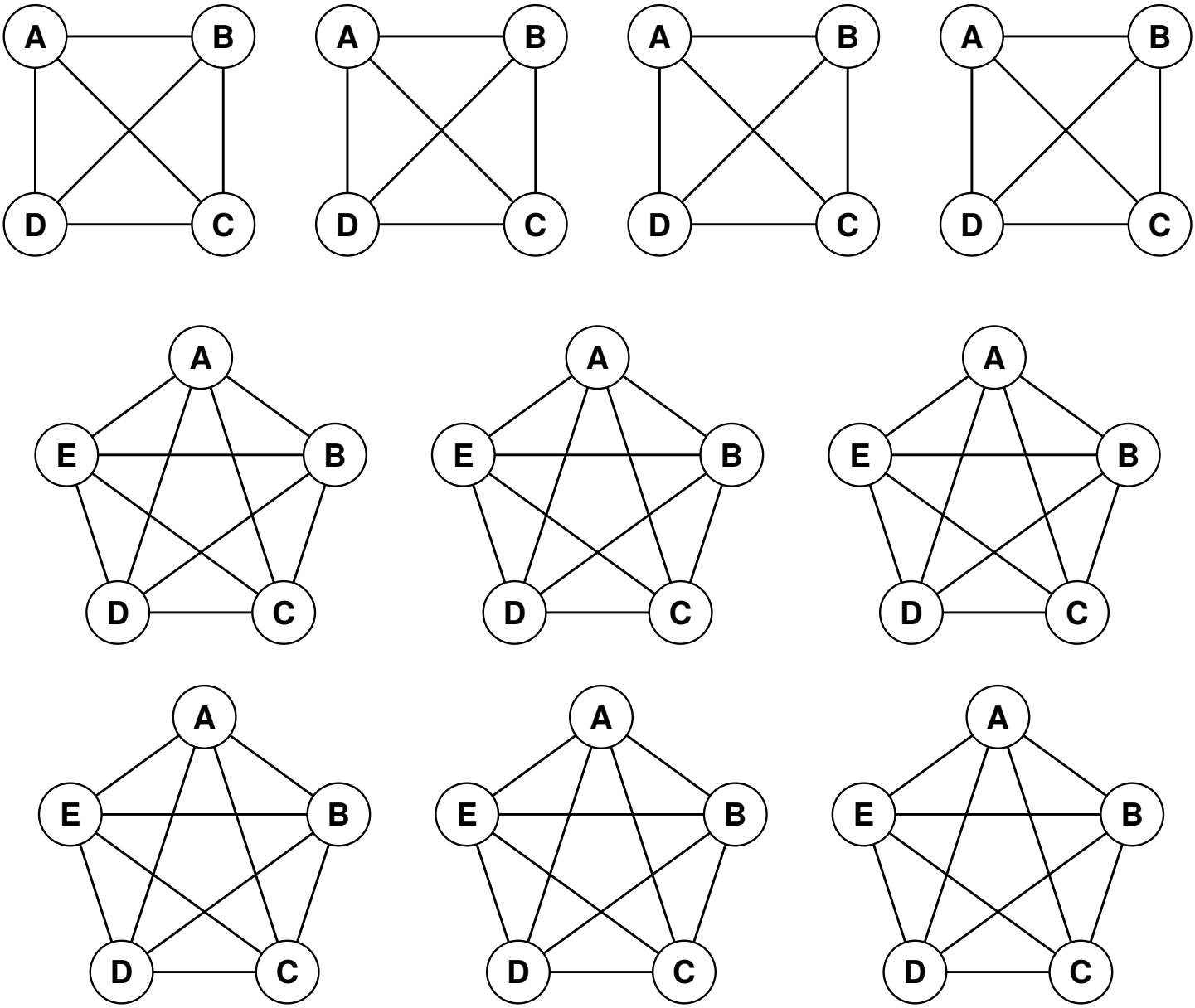






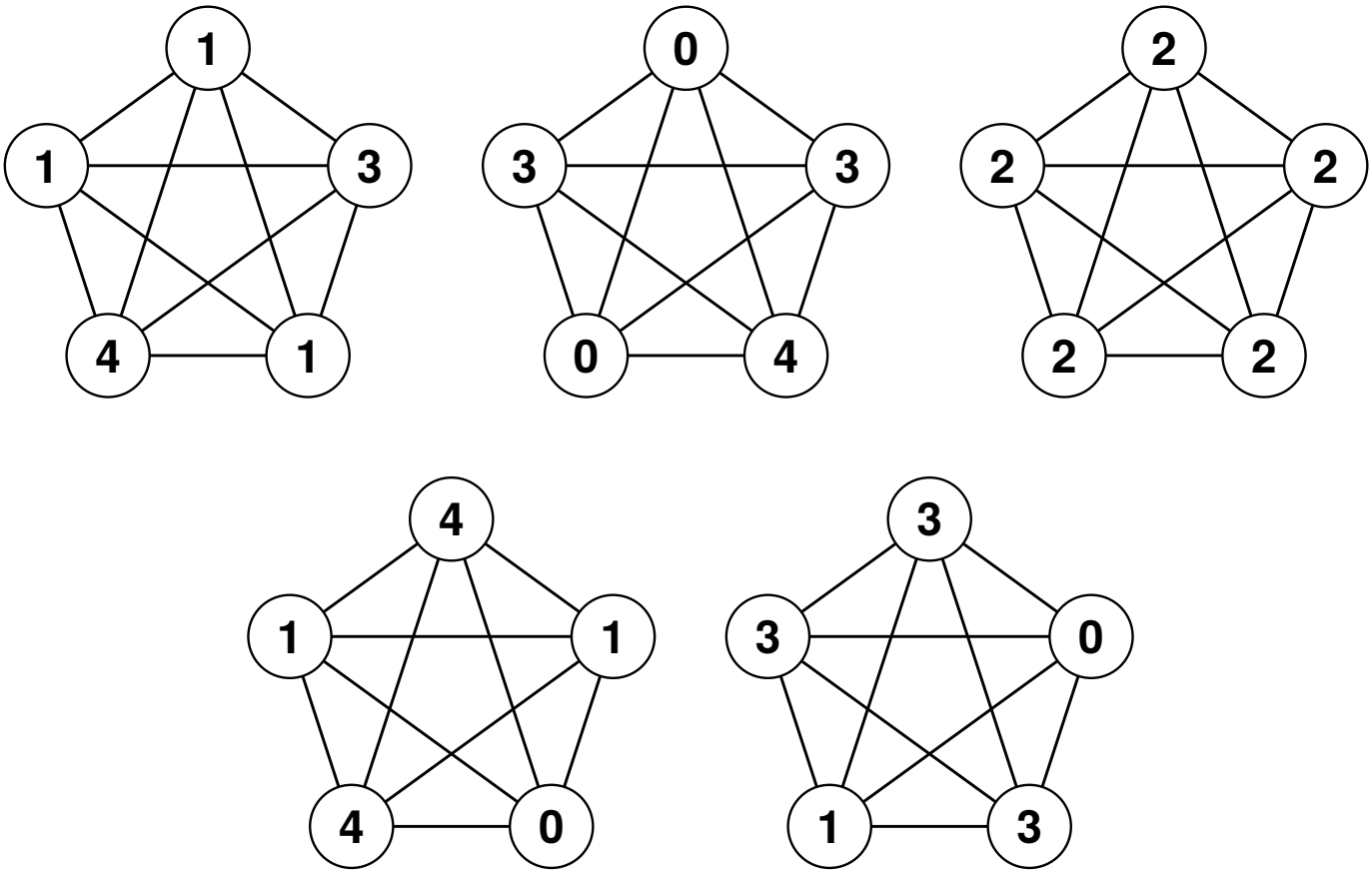
Problem 6: Is a player who won the most games always a king? Explain.

Problem 7: Which numbers of kings can a tournament of four players have? Which numbers can a tournament of five players have? Find an example of each number that can happen.

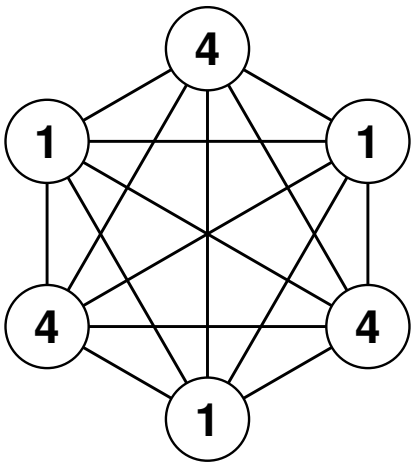


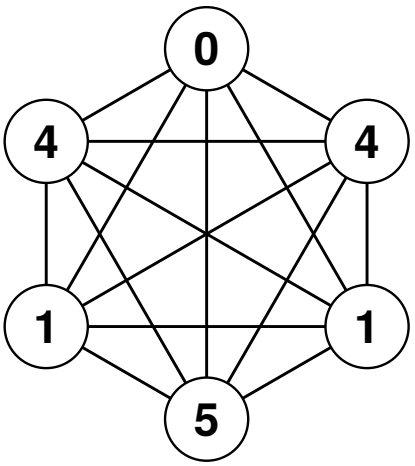
Problem 8: Can a tournament have exactly two kings? Explain.

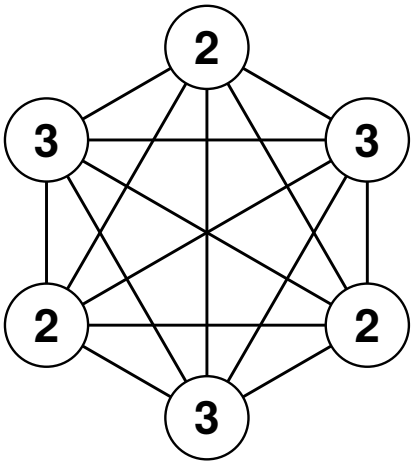
Problem 9: The number in each dot is how many games that player won. Draw arrowheads on the lines so that every player won exactly that many games, or explain why it cannot be done.

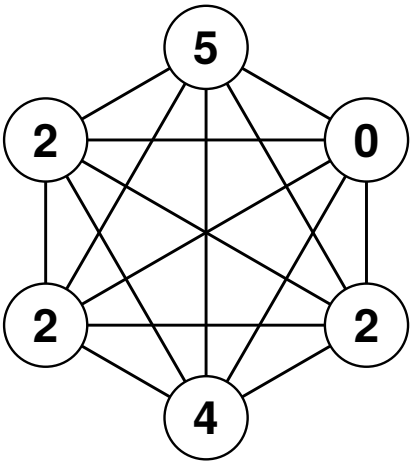


Problem 10: Do the same for six players.

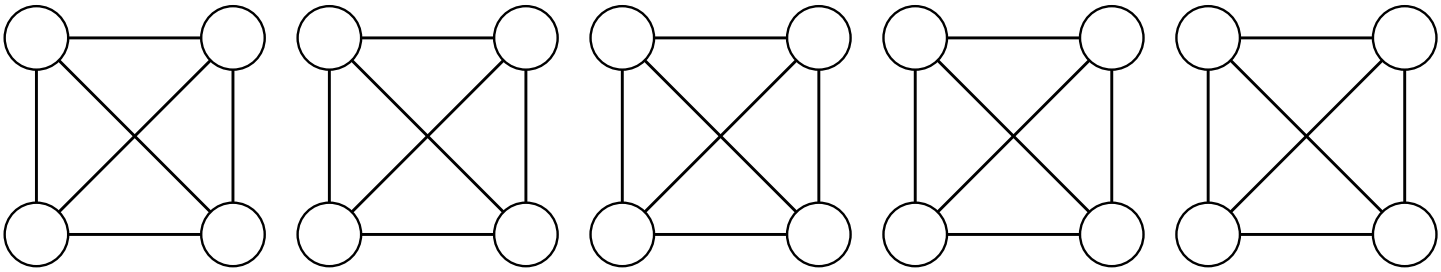


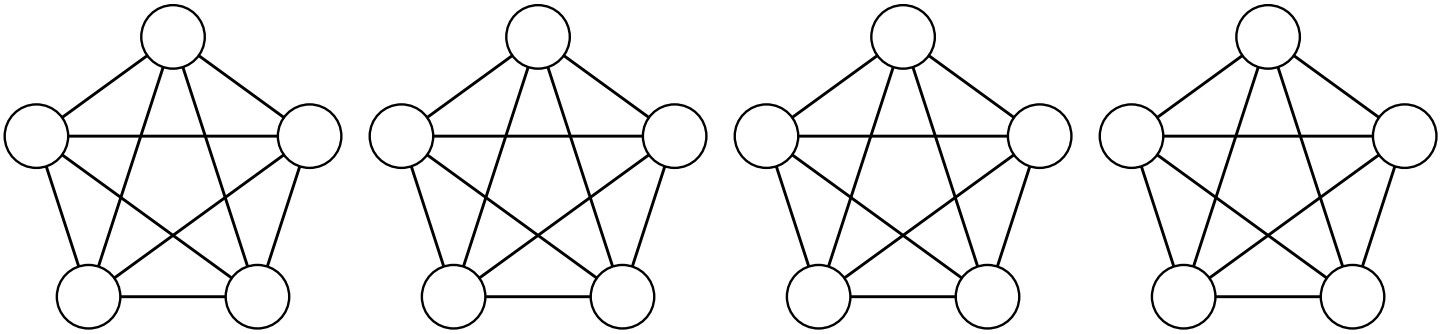


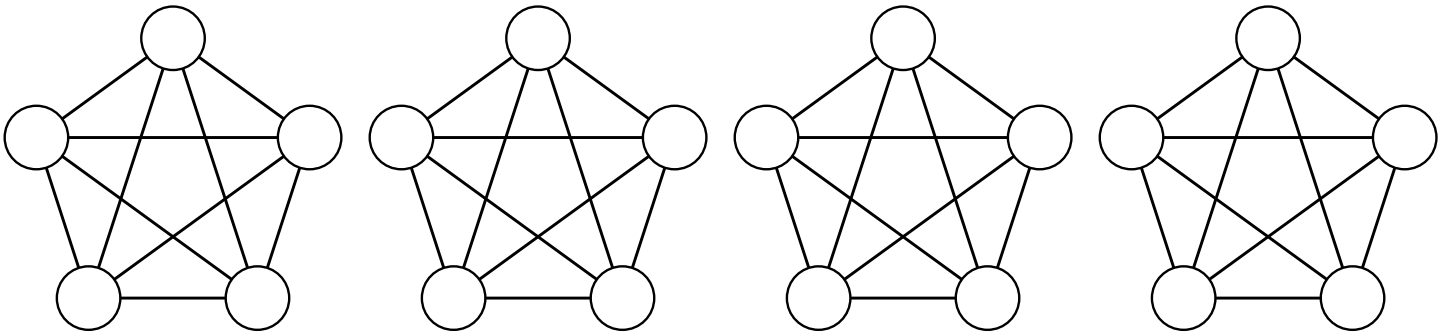


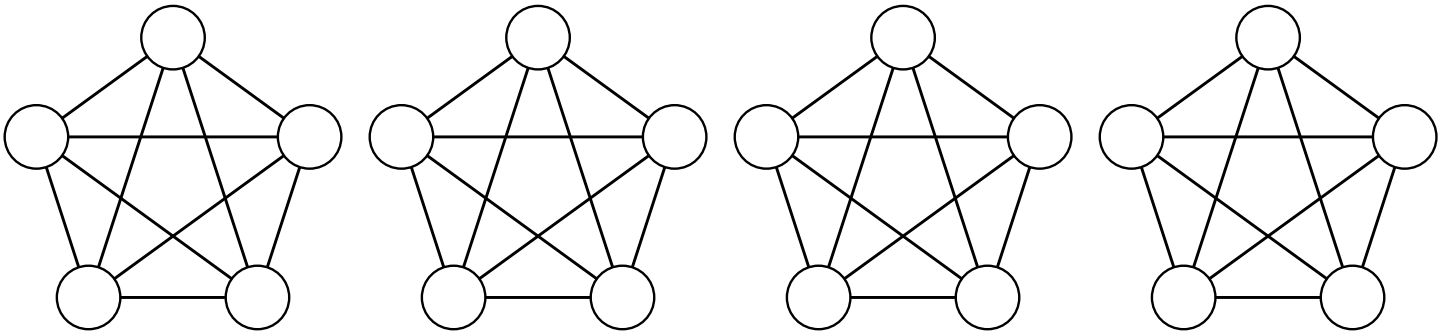


Problem 11: Find every list of win counts, written from smallest to largest, that can happen in a tournament of four players. Then do the same for five players.









Problem 12: Find a test that tells from a list of win counts alone whether some tournament has those win counts. Explain why a list that fails your test cannot happen.

Problem 13: Can a tournament of four players have exactly two line-ups? Can it have exactly four? Explain.